

SECURITY OPERATIONS

FIREWALL MANIPULATION: DETECTION & RESPONSE

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Introduction

This report documents a purple team exercise focused on Windows Firewall manipulation. Using MITRE Caldera as the adversary emulation platform, a firewall rule enumeration command was executed against a Windows 10 target and initially went undetected by the Wazuh SIEM. A custom detection rule was then engineered, tested, and validated, closing that gap.

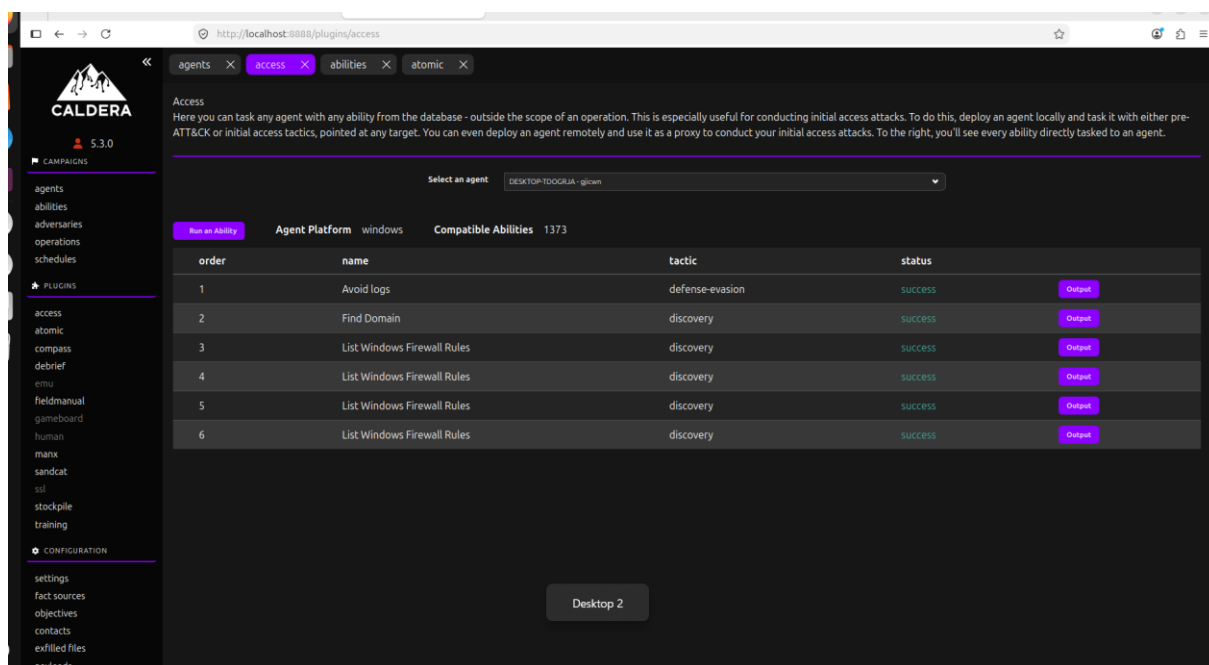
The same layered approach was applied to firewall modification activity, so that legitimate administrative changes could be distinguished from suspicious ones instead of the SIEM alerting on every netsh firewall command.

Environment & Scope

- Attacker platform: Kali Linux, running MITRE Caldera 5.3.0
- Target: Windows 10 host (DESKTOP-TDOGRJA)
- Detection stack: Wazuh SIEM, agent ID 001
- C2 framework: Caldera's Access plugin, used to task the agent with abilities outside a full operation

Attack Narrative

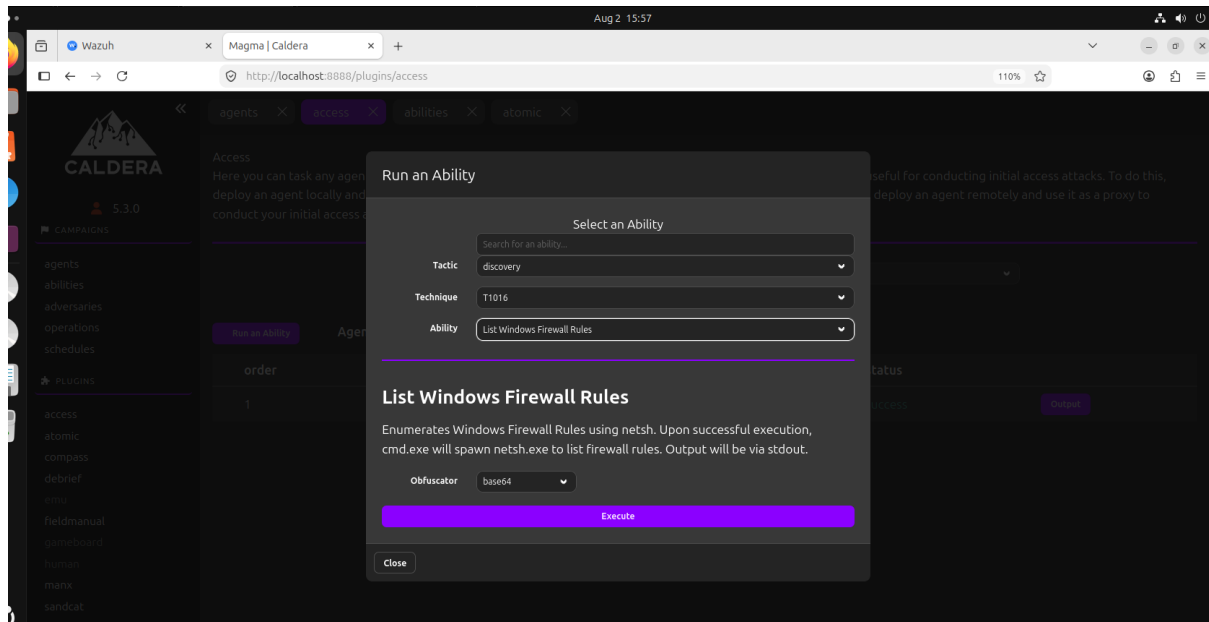
The attack chain ran three abilities in sequence through Caldera's Access plugin: Avoid Logs (defense evasion), Find Domain (discovery), and List Windows Firewall Rules (discovery, T1016), the last one run four times against the target agent.



The screenshot shows the MITRE Caldera web interface at <http://localhost:8080/plugins/access>. The interface includes a sidebar with navigation options like CAMPAIGNS, agents, abilities, adversaries, operations, schedules, and PLUGINS. The main content area is titled "Access" and provides instructions on how to use the plugin. Below the instructions, there is a "Select an agent" dropdown menu set to "DESKTOP-TDOGRJA: grom". A table displays a list of tasks, including "Avoid logs", "Find Domain", and "List Windows Firewall Rules", each with a status of "SUCCESS" and an "Output" button. The table also shows the "Agent Platform" as "windows" and "Compatible Abilities" as "1373".

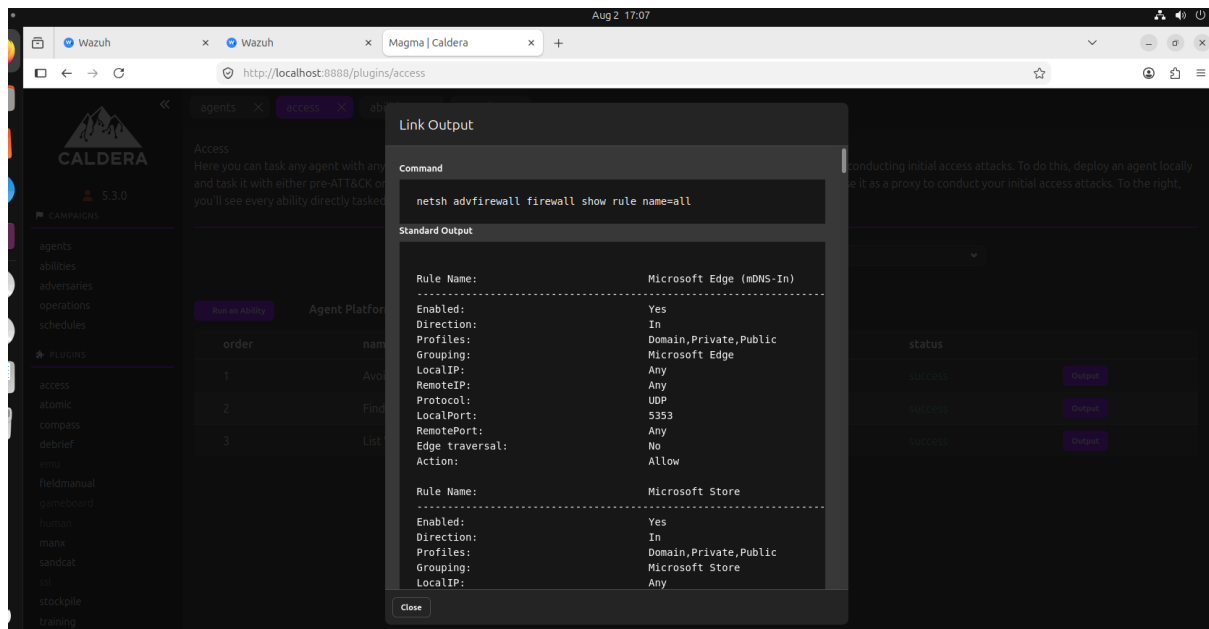
order	name	tactic	status
1	Avoid logs	defense-evasion	SUCCESS
2	Find Domain	discovery	SUCCESS
3	List Windows Firewall Rules	discovery	SUCCESS
4	List Windows Firewall Rules	discovery	SUCCESS
5	List Windows Firewall Rules	discovery	SUCCESS
6	List Windows Firewall Rules	discovery	SUCCESS

The firewall enumeration ability spawns cmd.exe, which in turn spawns netsh.exe to list all firewall rules, obfuscated with base64. Selecting the ability in Caldera shows its tactic and technique mapping before execution.



Execution succeeded. The command below returned the full local firewall rule set, confirming the attacker had complete visibility into the host's firewall posture.

```
netsh advfirewall firewall show rule name=all
```



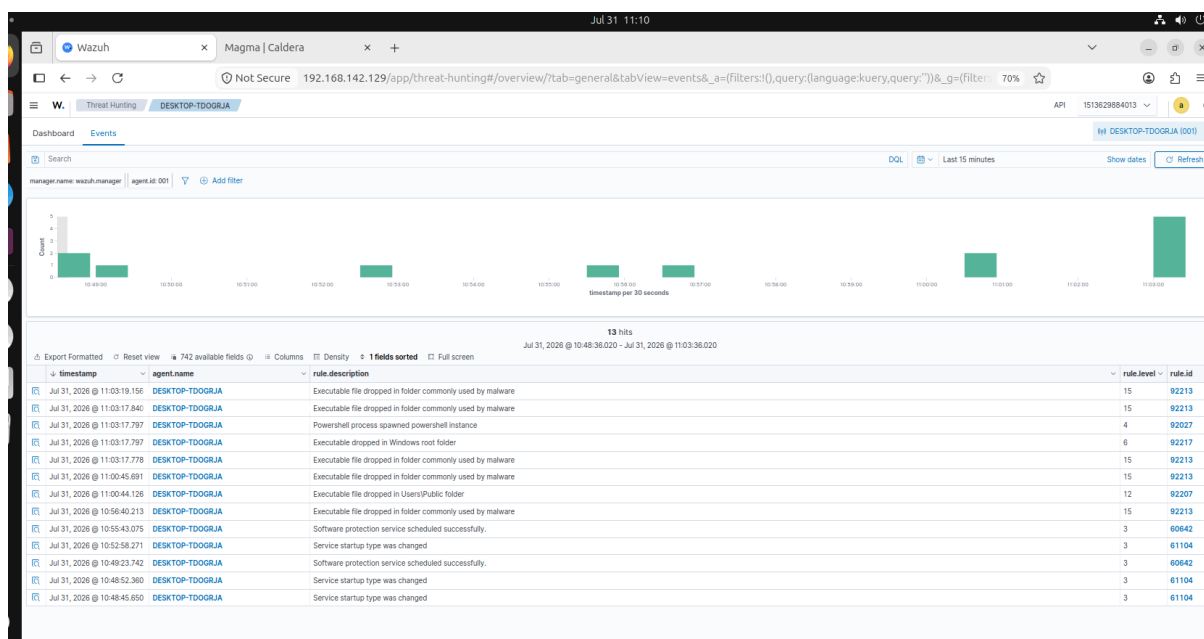
MITRE ATT&CK Mapping

Each stage of the exercise maps to a distinct tactic and technique, spanning both discovery and defense evasion:

Stage	Tactic	Technique	ID
Avoid Logs	Defense Evasion	Indicator Removal	T1070
Find Domain	Discovery	System Network Configuration Discovery	T1016
List Windows Firewall Rules	Discovery	System Network Configuration Discovery	T1016
Add / modify firewall rule (netsh)	Defense Evasion	Impair Defenses: Disable or Modify System Firewall	T1562.004

Detection Gap Analysis

Before the custom rule existed, Wazuh's threat hunting view showed activity in the surrounding time window, but none of it described the firewall enumeration itself. The alerts present were unrelated: executables dropped in folders commonly used by malware, a PowerShell process spawn, and service configuration changes. The firewall rule listing happened without triggering any dedicated detection.



Detection Engineering

Two separate rule chains were built, one for each side of the firewall activity.

Enumeration Detection

A single correlation rule (ID 100220, severity level 10) fires when a process spawns netsh with advfirewall firewall show arguments. This is what caught the discovery step above.

Modification Detection

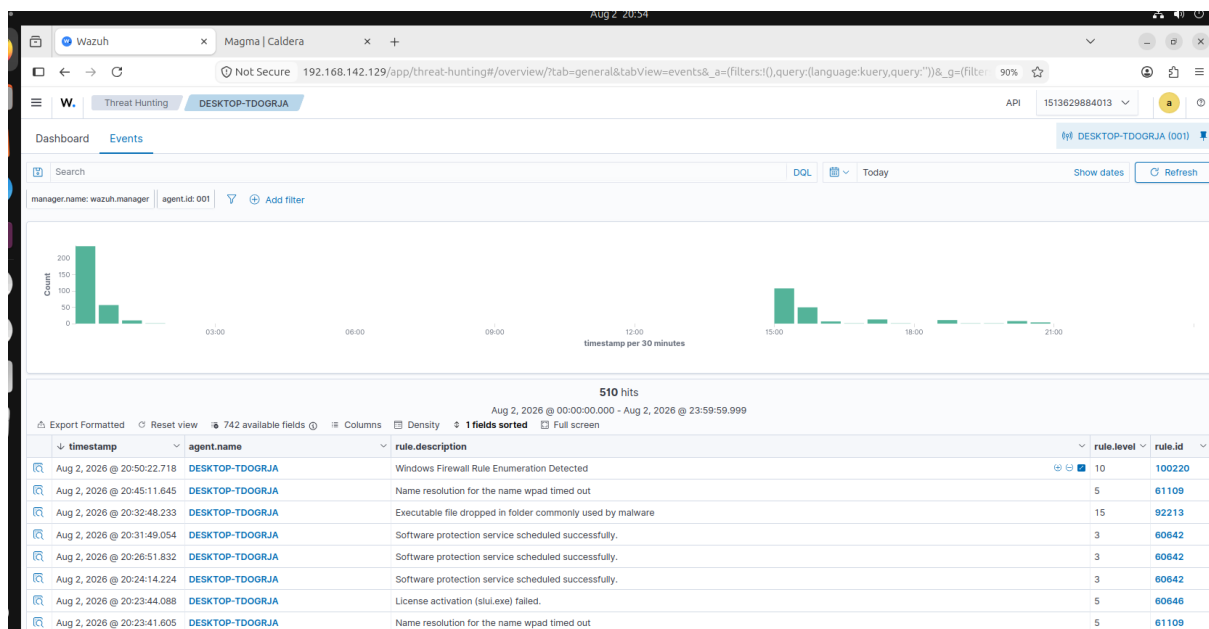
Modification detection is layered rather than a single blanket rule, since alerting on every netsh firewall command would be noisy in any environment where admins manage the firewall directly:

- Rule 92042 flags netsh.exe execution as a low-severity base event
- Rule 92043 escalates when that execution adds a firewall rule
- Rule 92044 escalates again when the added rule references local port 5900, associated with VNC

This staged approach means severity only climbs when the command pattern actually looks suspicious: first by confirming a rule was created, then by checking whether the specific port matches known remote-access tooling. Legitimate admin firewall changes and installers that create firewall exceptions during setup still generate the low-severity base event, but won't escalate unless the more specific conditions are met.

Validation

After the rule was deployed, the same enumeration technique was run again. “Windows Firewall Rule Enumeration Detected” now appears at the top of the Wazuh events view, severity level 10, rule ID 100220, correctly attributing the activity to firewall rule discovery.



Conclusion

The exercise closed a real detection gap: firewall rule enumeration was invisible to the SIEM until a purpose-built rule was written and validated against the same attack technique. Layering severity for the modification-side detections keeps the noise down without losing visibility into the specific patterns, like VNC-related ports, that matter most.